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Student Number

**2020**

# Mathematics Extension 2

## Trial HSC Examination

Date: Monday 10<sup>th</sup> August, 2020

### General

### Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

**Total Marks:**  
**100**

### Section I – 10 marks

- Allow about 15 minutes for this section

### Section II – 90 marks

- Allow about 2 hours and 45 minutes for this section

|                                  |                 |             |
|----------------------------------|-----------------|-------------|
| <b>Section I<br/>(10 marks)</b>  | Multiple Choice | /10         |
| <b>Section II<br/>(90 marks)</b> | Question 11     | /15         |
|                                  | Question 12     | /15         |
|                                  | Question 13     | /13         |
|                                  | Question 14     | /17         |
|                                  | Question 15     | /15         |
|                                  | Question 16     | /15         |
| <b>Total</b>                     |                 | <b>/100</b> |

***This question paper must not be removed from the examination room.***

*This assessment task constitutes 30% of the course.*

## Section I

10 marks

Allow about 15 minutes for this section

Use the multiple-choice sheet for Question 1–10

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1. If  $z = \frac{3+4i}{1+2i}$ , the imaginary part of  $z$  is:
- A.  $-2$
- B.  $-\frac{2}{5}$
- C.  $-2i$
- D.  $-\frac{2}{5}i$
2. If the vectors  $\underline{a} = m\underline{i} + 4\underline{j} + 3\underline{k}$  and  $\underline{b} = m\underline{i} + m\underline{j} - 4\underline{k}$  are perpendicular, then which of the following values of  $m$  are correct?
- A.  $m = -1$  or  $m = 1$
- B.  $m = -2$  or  $m = 0$
- C.  $m = -2$  or  $m = 6$
- D.  $m = -6$  or  $m = 2$
3. If  $z$  is any complex number satisfying  $|z - 1| = 1$ , then which of the following is correct?
- A.  $\arg(z - 1) = 2 \arg(z)$
- B.  $\arg(z - 1) = \arg(z + 1)$
- C.  $\arg(z) = 2 \arg(z + 1)$
- D.  $2 \arg(z) = \frac{2}{3} \arg(z^2 - z)$

Section I continues on the next page

4. A particle moves along a curve so that its position at time  $t$  is given by  $\vec{x} = \begin{pmatrix} t \\ \frac{1}{2}t^2 \\ \frac{1}{3}t^3 \end{pmatrix}$ .

The acceleration at  $t = 1$  is:

- A.  $\vec{j} + \vec{k}$   
 B.  $\vec{j} + 2\vec{k}$   
 C.  $2\vec{j} + \vec{k}$   
 D.  $\vec{i} + \vec{j} + 2\vec{k}$

5. A man walks a distance of 3 units from the origin in the direction of  $N45^\circ E$ , and then walks a distance of 4 units in the direction of  $N45^\circ W$  arriving at the point B. The position of B in the Argand plane is:

- A.  $3e^{\frac{i\pi}{4}} + 4i$   
 B.  $(3 - 4i)e^{\frac{i\pi}{4}}$   
 C.  $(3 + 4i)e^{\frac{i\pi}{4}}$   
 D.  $(4 + 3i)e^{\frac{i\pi}{4}}$

6. Using the substitution  $x = \pi - y$ , the definite integral  $\int_0^\pi x \sin x \, dx$  will simplify to:

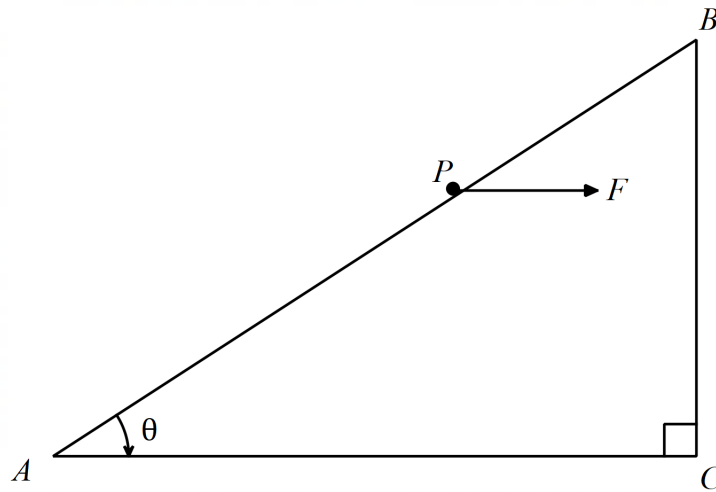
- A. 0  
 B.  $\frac{\pi^2}{4}$   
 C.  $\frac{\pi}{2} \int_0^\pi \sin x \, dx$   
 D.  $\int_0^\pi \sin x \, dx$

**Section I continues on the next page**

7. Which of the following statements is a negation of the following statement?

$$\forall x \in R^+, \exists y \in R^+ : xy = 1$$

- A.  $\exists x, y \in R^+$  such that  $xy \neq 1$
- B.  $\forall x, y \in R^+, \quad xy \neq 1$
- C.  $\exists x \in R^+ : \forall y \in R^+, xy \neq 1$
- D.  $\forall x \in R^+$ , there exists a positive real number  $y$  such that  $xy \neq 1$
8. A horizontal force  $F$  Newtons is applied to a small object  $P$  of mass  $m$  kg on a smooth plane, inclined to the horizontal at an angle  $\theta$ .



If  $F$  is just enough to keep  $P$  in equilibrium, then the magnitude of  $F$  is:

- A.  $mg \cos^2 \theta$
- B.  $mg \sin^2 \theta$
- C.  $mg \cos \theta$
- D.  $mg \tan \theta$

**Section I continues on the next page**

9. Which of the following is **false**?

A.  $\int_{-3}^3 x^3 e^{-x^2} dx = 0$

B.  $\int_{-4}^4 \frac{x^2}{x^2 + 4} dx = 2 \int_0^4 \frac{x^2}{x^2 + 4} dx$

C.  $\int_0^{\pi} \sin^4 \theta d\theta > \int_0^{\pi} \sin 4\theta d\theta$

D.  $\int_0^1 x^4 dx < \int_0^1 x^5 dx$

10.  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  are vectors of magnitude 3, 4 and 5 respectively.

It is also given that:

$\vec{a}$  is perpendicular to  $(\vec{b} + \vec{c})$ ,

$\vec{b}$  is perpendicular to  $(\vec{c} + \vec{a})$  and

$\vec{c}$  is perpendicular to  $(\vec{a} + \vec{b})$ .

Then, the magnitude of the vector  $\vec{a} + \vec{b} + \vec{c}$  is:

A. 5

B.  $5\sqrt{2}$

C.  $5\sqrt{3}$

D. 12

**End of Section I**

## Section II

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your response should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use the Question 11 Writing Booklet.

- (a) Let  $\omega_1 = 8 - 2i$  and  $\omega_2 = -5 + 3i$ . 2  
Find:

$$\omega_1 + \overline{\omega_2}$$

- (b) (i) Express  $z = \sqrt{2} - i\sqrt{2}$  in the exponential form 2  
(ii) Hence, write  $z^{22}$  in the  $a + ib$  form where  $a, b \in R$  2

- (c) (i) Find the square roots of  $-35 + 12i$  3  
(ii) Solve  $z^2 - (5 + 4i)z + 11 + 7i = 0$  2

- (d) Find 2

$$\int \frac{dx}{x(\ln x)^2}$$

- (e) Find 2

$$\int \frac{1}{x^2 - 6x + 13} dx$$

**End of Question 11**

**Question 12** (15 marks) Use the Question 12 Writing Booklet.

- (a) Prove the following statement using a proof by contradiction.

4

“For each irrational number  $s$ , the number  $2s + 1$  is also irrational”

- (b) Show that

4

$$\int_1^3 \frac{6t + 23}{(2t - 1)(t + 6)} dt = \ln \frac{225}{7}$$

- (c) Evaluate

3

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos\theta}$$

by using the substitution  $t = \tan \frac{\theta}{2}$

- (d) (i) Show that if  $1, \omega_1, \omega_2$  are the cube roots of 1,

1

$$1 + \omega_1 + \omega_1^2 = 0$$

and

$$1 + \omega_2 + \omega_2^2 = 0$$

- (ii) If  $n$  is not a multiple of 3, prove that

3

$$x^{2n} + x^n + 1 \text{ is divisible by } x^2 + x + 1$$

**End of Question 12**

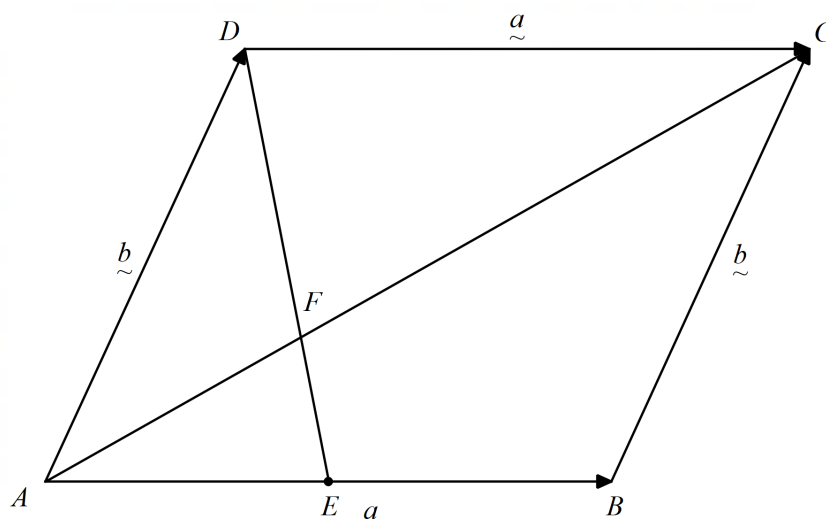
**Question 13** (13 marks) Use the Question 13 Writing Booklet.

- (a) Prove by mathematical induction that  $\forall n \in \mathbb{Z}^+$

3

$$\sum_{i=1}^n \sqrt{i} > \frac{2n\sqrt{n}}{3}$$

- (b)  $ABCD$  is a parallelogram and  $E$  is the midpoint of  $AB$ .



Using vectors, show that any line joining any vertex of a parallelogram to the midpoint of a side not passing through that vertex divides the opposite diagonal in the ratio 1:2. (i.e. Show  $DE$  divides  $AC$ , in the ratio 1:2)

4

- (c) The acceleration of a particle moving along the  $x$ -axis is given by

$$\frac{d^2x}{dt^2} = 2x^3 - 10x$$

- (i) If the particle starts at the origin with velocity  $u$ , show that its velocity is given by  $v^2 - u^2 = x^4 - 10x^2$

2

- (ii) If  $u = 3$ , show that the particle oscillates within the interval  $-1 \leq x \leq 1$

3

- (iii) Is the motion referred to in (ii), an example of simple harmonic motion? Give clear reason for your answer.

1

**End of Question 13**

**Question 14** (17 marks) Use the Question 14 Writing Booklet.

(a) (i) Show that  $a^2 + b^2 > 2ab$ , where  $a$  and  $b$  are distinct positive real numbers 1

(ii) Hence, show that  $a^2 + b^2 + c^2 > ab + bc + ca$ , where  $a, b$  and  $c$  are distinct positive real numbers 2

(iii) Hence, or otherwise prove that 2

$$\frac{a^2b^2 + b^2c^2 + c^2a^2}{a + b + c} > abc,$$

Where  $a, b$  and  $c$  are distinct positive real numbers

(b) (i) If  $Z = \cos \theta + i \sin \theta$ , 1  
Express  $Z^n + Z^{-n}$  in terms of  $\theta$

Let, for real values of  $k$ ,

$$f(\theta) = 1 + k \cos \theta + k^2 \cos 2\theta + k^3 \cos 3\theta + \cdots + k^n \cos n\theta + \cdots$$

(ii) Using the result in (i) and expressing  $f(\theta)$  as the sum of two geometric progressions, prove that 3

$$f(\theta) = \frac{1 - k \cos \theta}{1 - 2k \cos \theta + k^2}, \quad |k| < 1$$

(iii) Verify the result in (ii) for  $|k| < 1$  and  $\theta = \frac{\pi}{2}$  1

**Question 14 continues on the next page**

**Question 14 (continued)**

- (c)  $\tilde{r}_1$  and  $\tilde{r}_2$  are two lines with vector equations:

$$\tilde{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \text{ and}$$

$$\tilde{r}_2 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \quad \lambda, \mu \in R$$

- (i) Show that these two lines intersect. **3**
- (ii) Find the angle between the lines. **1**
- (iii) Find the shortest distance from the point  $P(1,2,0)$  to the line **3**

$$\tilde{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

**End of Question 14**

**Question 15** (15 marks) Use the Question 15 Writing Booklet.

- (a) If the functions  $f(x)$  and  $g(x)$  are such that  $f(x) > g(x) \geq 0$  for  $a \leq x \leq b$ , by using a sketch (or otherwise) explain why 2

$$\int_a^b f(x) dx > \int_a^b g(x) dx$$

- (b) Let

$$I_n = \int_0^1 (1 - t^2)^{\frac{n-1}{2}} dt,$$

Where,  $n$  is a non-negative integer.

- (i) Using integration by parts, or otherwise, show that 4

$$n I_n = (n - 1) I_{n-2} \quad \text{if } n \geq 2$$

- (ii) Let  $J_n = n I_n I_{n-1}$ ,  $n \geq 1$ . 4

Show that

$$J_n = \frac{\pi}{2}, \quad \forall n \geq 1$$

- (iii) Using part (a), or otherwise, show that 3

$$0 < I_n < I_{n-1}$$

- (iv) Hence, or otherwise, prove that 2

$$\sqrt{\frac{\pi}{2n+2}} < I_n < \sqrt{\frac{\pi}{2n}}$$

**End of Question 15**

**Question 16** (15 marks) Use the Question 16 Writing Booklet.

- (a) When a jet aircraft touches down, two different retarding forces combine to bring it to rest. If the jet has a mass of  $M$  kg and a speed of  $v$  m/s, there is a constant frictional force of  $\frac{1}{4}M$  newtons and a force of  $\frac{1}{108}Mv^2$  newtons due to the reverse thrust of the engines.

The reverse thrust of the engines do not take into effect until 20 seconds after touch down.

- (i) Show that

2

$$\frac{d^2x}{dt^2} = -\frac{1}{4} \quad \text{for } 0 < t \leq 20$$

And that for  $t > 20$ , and until after the jet stops,

$$\frac{d^2x}{dt^2} = -\frac{1}{108}(27 + v^2)$$

- (ii) If the jet's speed at touch down is 60 m/s, show that  $v = 55$  and  $x = 1150$  at the instant the reverse thrust of the engines takes effect.

2

- (iii) Show that when  $t > 20$ ,

2

$$x = 1150 + 54 \{\ln(27 + 55^2) - \ln(27 + v^2)\}$$

- (iv) Calculate how far from the touchdown point the jet comes to rest. Give your answer to the nearest metre.

1

**Question 16 continues on the next page**

(b) By considering the sum to  $n$  terms of the series

$$1 - t + t^2 - t^3 + \cdots + (-1)^{n-1}t^{n-1}, \quad t \neq -1$$

(i) Show that

2

$$x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots + (-1)^n \frac{x^n}{n} = \log_e(1+x) - (-1)^n \int_0^x \frac{t^n}{1+t} dt$$

for  $0 < t < x$ ,

(ii) Also, show that

2

$$\int_0^x \frac{t^n}{1+t} dt < \int_0^x t^n dt$$

(iii) For  $0 < x \leq 1$ , show that

2

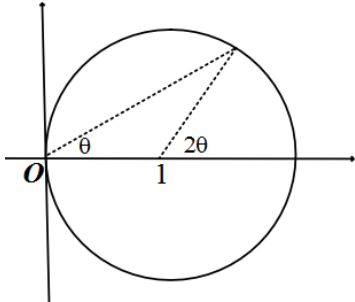
$$\int_0^x \frac{t^n}{1+t} dt < \frac{1}{n+1}$$

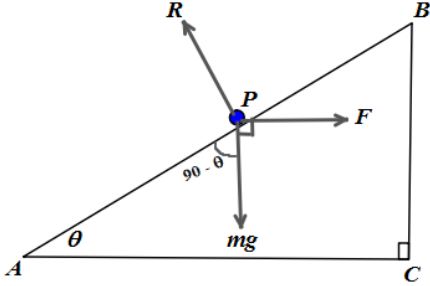
(iv) Hence, prove that as  $n \rightarrow \infty$ ,

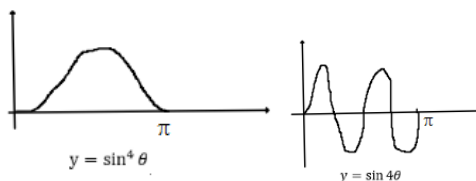
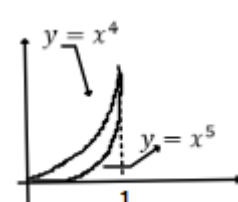
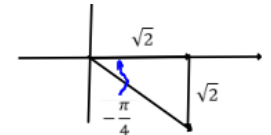
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$$x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots + (-1)^n \frac{x^n}{n} + \cdots = \log_e(1+x)$$

**End of Examination**

| Multiple Choice |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |  |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 1.              | $z = \frac{3+4i}{1+2i}$ $= \frac{(3+4i)(1-2i)}{(1+2i)(1-2i)}$ $= \frac{11}{5} + \left(-\frac{2}{5}\right)i$ $\text{Im}(z) = -\frac{2}{5}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | B |  |
| 2.              | $\vec{a} = m\vec{i} + 4\vec{j} + 3\vec{k} \text{ and}$ $\vec{b} = m\vec{i} + m\vec{j} - 4\vec{k}$ <p>If <math>\vec{a}</math> and <math>\vec{b}</math> are perpendicular, <math>\vec{a} \cdot \vec{b} = 0</math></p> $m^2 + 4m - 12 = 0$ $(m + 6)(m - 2) = 0$ $m = -6, 2$                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D |  |
| 3.              |  <p>Angle subtended at the centre is double the angle at the circumference</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | A |  |
| 4.              | $\vec{r} = \begin{pmatrix} t \\ \frac{1}{2}t^2 \\ \frac{1}{3}t^3 \end{pmatrix}; \quad \vec{\dot{r}} = \begin{pmatrix} 1 \\ t \\ t^2 \end{pmatrix}; \quad \vec{\ddot{r}} = \begin{pmatrix} 0 \\ 1 \\ 2t \end{pmatrix}$ <p>At <math>t = 1</math>, <math>\vec{\ddot{r}} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \vec{j} + 2\vec{k}</math></p>                                                                                                                                                                                                                                                                                                                                                                          | B |  |
| 5.              | <p>Suppose the man reaches A after walking 3 units in <math>N45^\circ E</math> and B after walking a distance of 4 units in <math>N45^\circ W</math>. Position of A in the Argand diagram is <math>3e^{\frac{i\pi}{4}}</math>. Let the position of B be <math>z</math>. Since <math>\angle OAB = \frac{\pi}{2}</math>, we have</p> $\arg\left(\frac{0 - 3e^{\frac{i\pi}{4}}}{z - 3e^{\frac{i\pi}{4}}}\right) = \frac{\pi}{2}$ $\frac{0 - 3e^{\frac{i\pi}{4}}}{z - 3e^{\frac{i\pi}{4}}} = \frac{OA}{AB} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$ $\frac{0 - 3e^{\frac{i\pi}{4}}}{z - 3e^{\frac{i\pi}{4}}} = \frac{3i}{4}$ $z - 3e^{\frac{i\pi}{4}} = 4ie^{\frac{i\pi}{4}}$ $z = (3 + 4i)e^{\frac{i\pi}{4}}$ | C |  |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |  |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| 6. | $\int_0^{\pi} x \sin x \, dx = \int_0^{\pi} x \sin x \, dx$ <p> <math>x = \pi - y</math><br/> <math>dx = -dy</math><br/> <math>x = 0, y = \pi</math><br/> <math>x = \pi, y = 0</math> </p> $\int_0^{\pi} x \sin x \, dx = \int_{\pi}^0 (\pi - y) \sin(\pi - y) \times -dy$ $= \int_0^{\pi} (\pi - y) \sin y \, dy$ $= \int_0^{\pi} \pi \sin y \, dy - \int_0^{\pi} y \sin y \, dy$ $\int_0^{\pi} x \sin x \, dx = \int_0^{\pi} y \sin y \, dy$ $2 \int_0^{\pi} y \sin y \, dy = \pi \int_0^{\pi} \sin y \, dy$ $\int_0^{\pi} y \sin y \, dy = \frac{\pi}{2} \int_0^{\pi} \sin y \, dy$ | C |  |
| 7. | C<br>$\exists x \in R^+ : \forall y \in R^+, xy \neq 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | C |  |
| 8. |  <p>By Lami's Theorem,</p> $\frac{R}{\sin 90^\circ} = \frac{F}{\sin(180^\circ - \theta)} = \frac{mg}{\sin(90^\circ + \theta)}$ $\frac{R}{1} = \frac{F}{\sin \theta} = \frac{mg}{\cos \theta}$ $F = mg \tan \theta$                                                                                                                                                                                                                                                                                  | D |  |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                        |           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 9.          | <p>(A) <math>\int_{-3}^3 x^3 e^{-x^2} dx = 0</math> odd function, True</p> <p>(B) Even function, True</p> <p>(C)</p>  <p>(D)</p>  <p>True</p> <p>False</p>                                                                                                                                                                                                                                                                                | D                                                                                                                                      |           |
| 10.         | $\begin{aligned}  \vec{a} + \vec{b} + \vec{c} ^2 &= (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) \\ &= \vec{a} \cdot \vec{a} + \vec{a} \cdot (\vec{b} + \vec{c}) + \vec{b} \cdot \vec{b} + \vec{b} \cdot (\vec{a} + \vec{c}) + \\ &\quad \vec{c} \cdot \vec{c} + \vec{c} \cdot (\vec{a} + \vec{b}) \\ &=  \vec{a} ^2 + \vec{a} \cdot (\vec{b} + \vec{c}) +  \vec{b} ^2 + \vec{b} \cdot (\vec{a} + \vec{c}) + \\ &\quad  \vec{c} ^2 + \vec{c} \cdot (\vec{a} + \vec{b}) \\ &= 9 + 0 + 16 + 0 + 25 + 0 \\ &= 50 \\  \vec{a} + \vec{b} + \vec{c}  &= \sqrt{50} = 5\sqrt{2} \end{aligned}$ | B                                                                                                                                      |           |
| Question 11 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                        |           |
| (a)         | $\omega_1 = 8 - 2i$ and $\omega_2 = -5 + 3i$<br>$\omega_1 = 8 - 2i$ and<br>$w_1 + \overline{w_2} = 8 - 2i - 5 - 3i$<br>$3 - 5i$                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>1 mark: correct <math>\overline{w_2}</math></p> <p>1 mark: correctly adds <math>\omega_1</math> and <math>\overline{w_2}</math></p> |           |
| (b)(i)      |  $\begin{aligned} z &= \sqrt{2} - i\sqrt{2} \\  z  &= \sqrt{2 + 2} = 2 \\ \arg(z) &= -\frac{\pi}{4} \\ z &= 2 e^{-\frac{i\pi}{4}} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                         | <p>1 mark: correct <math> z </math> or correct <math>\arg(z)</math></p> <p>1 mark: correct exponential form</p>                        | Well done |

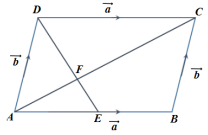
|         |                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                               |                                                                                   |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| (ii)    | $z^{22} = 2^{22} e^{\frac{-i22\pi}{4}}$ $z^{22} = 2^{22} e^{\frac{-i11\pi}{2}}$ $z^{22} = 2^{22} \left( \cos\left(-\frac{11\pi}{2}\right) + i\sin\left(-\frac{11\pi}{2}\right) \right)$ $z^{22} = 2^{22} \left( \cos\left(\frac{11\pi}{2}\right) - i\sin\left(\frac{11\pi}{2}\right) \right)$ $z^{22} = 2^{22}(0 - (-i))$ $z^{22} = 2^{22}i$   | <p>1 mark: correctly writes the expression for <math>z^{22}</math></p> <p>1 mark: correctly evaluates and gives the answer in the simplest form</p>                                           | <p>Some minor in calculation of <math>\sin\left(\frac{11\pi}{2}\right)</math></p> |
| (c)(i)  | <p>Let <math>\sqrt{-35 + 12i} = a + ib</math></p> $a^2 - b^2 = -35$ $a^2 + b^2 = \sqrt{(-35)^2 + 12^2} = 37$ $a^2 = 1 \quad \therefore a = \pm 1$ $b^2 = 36 \quad \therefore b = \pm 6$ $\sqrt{-35 + 12i} = \pm(1 + 6i)$                                                                                                                       | <p>1 mark: gives the correct values for <math>a^2 - b^2</math> and <math>2ab = 12</math></p> <p>2 mark: correctly evaluates a and b and gives the square root.</p> <p>1 mark: Minor error</p> | <p>Well done (multiple methods used)</p>                                          |
| (c)(ii) | $z^2 - (5 + 4i)z + 11 + 7i = 0$ $z = \frac{(5 + 4i) \pm \sqrt{(5 + 4i)^2 - 4(11 + 7i)}}{2}$ $z = \frac{(5 + 4i) \pm \sqrt{-35 + 12i}}{2}$ <p>Using (c)(i),</p> $z = \frac{(5 + 4i) \pm (1 + 6i)}{2}$ $z = \frac{5 + 4i + 1 + 6i}{2}, \frac{5 + 4i - 1 - 6i}{2}$ $z = \frac{5 + 4i + 1 - 6i}{2}, \frac{5 + 4i - 1 + 6i}{2}$ $z = 3 - i, 2 + 5i$ | <p>1 mark: applies quadratic formula correctly</p> <p>1 mark: Uses the solution from (c)(i) to give all the correct roots</p>                                                                 | <p>Well done (other than careless errors)</p>                                     |
| (d)     | $I = \int \frac{dx}{x(\ln x)^2}$ <p>Let <math>u = \ln x</math></p> $\frac{du}{dx} = \frac{1}{x}$ $du = \frac{1}{x} dx$ $I = \int \frac{du}{u^2}$ $= -\frac{1}{u} + c$ $= -\frac{1}{\ln x} + c$                                                                                                                                                 | <p>2 mark: Correct answer from correct working</p> <p>1 mark: Minor error</p>                                                                                                                 | <p>Well done</p>                                                                  |
| (e)     | $\int \frac{1}{x^2 - 6x + 13} dx$ $\int \frac{1}{(x - 3)^2 + 4} dx$ $= \frac{1}{2} \tan^{-1} \frac{x - 3}{2} + C$                                                                                                                                                                                                                              | <p>2 mark: Correct answer from correct working</p> <p>1 mark: Minor error</p>                                                                                                                 | <p>Well done</p>                                                                  |

| Question 12 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)         | <p>The statement to prove is:<br/> <math>\forall s \in R, \text{ if } s \notin Q, \text{ then } 2s + 1 \notin Q</math></p> <p>Proof:<br/>         Suppose the statement is false.<br/>         That is <math>\exists s \in R : s \notin Q</math> and <math>2s + 1 \in Q</math>. 1 mark</p> <p>Then,</p> $2s + 1 = \frac{a}{b} \text{ for some } a, b \in J \text{ and } b \neq 0.$ $2s = \frac{a}{b} - 1 = \frac{a - b}{b}$ $s = \frac{a - b}{2b}$ $\therefore s = \frac{c}{d} \text{ for some } c, d \in J \text{ and } d \neq 0.$ <p>However, <math>s \notin Q</math>. Thus we have reached a contradiction in our assumption that the original statement was false.<br/>         Thus the statement is true.</p> | <p>1 mark: Gives proof with correct notation and fluid logic- must demonstrate mastery of mathematical notation</p> <p>1 mark: Writes the statement of contradiction</p> <p>2 marks: for correct proof (demonstrates the contradiction and states the meaning of each statement.</p> <p>1 mark: Gives the correct proof with not-so-good explanation</p> | <p>Poorly done.</p> <p>Try to uses as much mathematical notation as possible. (rewrite the given statement using mathematical notation)</p> <p>Please see the contradiction statement: most students <math>\forall s</math> instead of <math>\exists s \in R</math></p> |
| (b)         | $\int_1^3 \frac{6t + 23}{(2t - 1)(t + 6)} dt$ $\frac{6t + 23}{(2t - 1)(t + 6)} = \frac{A}{2t - 1} + \frac{B}{t + 6}$ $A = 4, B = 1$ $\int_1^3 \frac{6t + 23}{(2t - 1)(t + 6)} dt = \int_1^3 \frac{4}{2t - 1} + \frac{1}{t + 6} dt$ $= \int_1^3 \frac{2 \times 2}{2t - 1} + \frac{1}{t + 6} dt$ $[\ln((2t - 1)^2(t + 6))]_1^3$ $= \ln \frac{25 \times 9}{1 \times 7}$ $= \ln \frac{225}{7}$                                                                                                                                                                                                                                                                                                                          | <p>1 mark: correctly converts to partial fractions</p> <p>2 mark: correct integration (1 mark: Minor error in integration)</p> <p>1 mark: correct evaluation</p>                                                                                                                                                                                         | Well done                                                                                                                                                                                                                                                               |
| (c)         | $\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos\theta}$ $t = \tan \frac{\theta}{2}$ $dt = \frac{1}{2} \sec^2 \frac{\theta}{2} dt = (1 + t)^2 dt$ $\theta = 0, t = 0$ $\theta = \frac{\pi}{2}, t = 1$ <p style="text-align: right;"><b>1 mark</b></p> $\int_0^1 \frac{2}{2 + \frac{1 - t^2}{1 + t^2}} dt$ $\int_0^1 \frac{2dt}{2(1 + t^2) + 1 - t^2}$ $\int_0^1 \frac{2dt}{3 + t^2}$ <p style="text-align: right;"><b>1 mark</b></p> $\left[ \frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$ $\frac{2}{\sqrt{3}} \times \left( \frac{\pi}{6} - 0 \right)$ $\frac{\pi}{3\sqrt{3}}$ <p style="text-align: right;">1 mark:</p>                                                                             | <p>1 mark: correctly converts the limits and <math>d\theta</math> in terms t</p> <p>1 mark: correctly converts the integrand</p> <p>2 marks: correctly integrates and evaluates</p> <p>1 mark: minor error in integration and evaluation</p>                                                                                                             | Well done, except for error in simplification of the integrand and making the converted from far too simple                                                                                                                                                             |

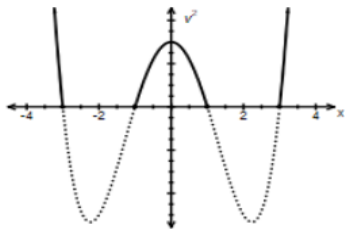
|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| (d)(i) | <p>Since, <math>\omega_1</math> is a cube root of unity, <math>\omega_1^3 - 1 = 0</math>.<br/> ie. <math>(\omega_1 - 1)(\omega_1^2 + \omega_1 + 1) = 0</math>.<br/> As <math>\omega_1 \neq 0</math>,<br/> <math display="block">(\omega_1^2 + \omega_1 + 1) = 0</math></p> <p>Also,<br/> <math display="block">(\omega_2^2 + \omega_2 + 1) = 0</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 mark: factorises $\omega_1^3 - 1 = 0$ and explains the results with correct reasons.                                                                         | Well done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (ii)   | <p>Let <math>\alpha = \omega_1^n</math>. Then <math>\alpha \neq 1</math> and <math>\alpha^3 = (\omega_1^n)^3 = (\omega_1^3)^n = 1</math></p> <p><math>\therefore \alpha</math> is a cube root of 1. <b>1 mark</b></p> <p>By (a), <math>\alpha^2 + \alpha + 1 = 0</math></p> <p>Thus <math>\omega_1^{2n} + \omega_1^n + 1 = 0</math></p> <p>ie. <math>\omega_1</math> is a root of <math>x^{2n} + x^n + 1</math></p> <p>Similarly if <math>\beta = \omega_2^n</math>, <math>\beta^2 + \beta + 1 = 0</math>,<br/> <math display="block">\omega_2^{2n} + \omega_2^n + 1 = 0</math></p> <p><math>\omega_2</math> is a root of <math>x^{2n} + x^n + 1</math></p> <p>From (i), using remainder theorem, <span style="float: right;">1</span><br/> <math display="block">x^2 + x + 1 = (x - \omega_1)(x - \omega_2)</math></p> <p>But we have</p> <p><math>\omega_1^{2n} + \omega_1^n + 1 = 0</math> and <math>\omega_2^{2n} + \omega_2^n + 1 = 0</math> <span style="float: right;">2</span><br/> <math>\omega_1</math> and <math>\omega_2</math> are roots of <math>x^{2n} + x^n + 1 = 0</math></p> <p><math>\therefore (x - \omega_1)</math> and <math>(x - \omega_2)</math> are factors of <span style="float: right;">3</span><br/> <math>x^{2n} + x^n + 1</math></p> <p>Thus,<br/> <math>x^{2n} + x^n + 1</math> is divisible by <math>x^2 + x + 1</math> <span style="float: right;">4</span></p> | <p>1 mark: proves <math>\omega^n</math> is a root of <math>x^3 - 1 = 0</math></p> <p>1 mark: explains <math>\alpha</math> and <math>\beta</math> are roots</p> | <p>Poorly done.</p> <p>Many students used the method of representing <math>n = 3k + 1</math> or <math>n = 3k + 2</math> which is fine.</p> <p>However, when you substitute <math>\omega_1</math> into <math>x^{2(3k+1)} + x^{(3k+1)} + 1</math> and prove that it satisfies the equation as <math>(\omega_1^2 + \omega_1 + 1) = 0</math>, you are only proving <math>\omega_1</math> is a root.</p> <p>Statements 1, 2, 3 and 4 must be given to complete the proof</p> <p>The question is about proving <math>\omega_1</math> and <math>\omega_2</math> are roots of <math>x^{2n} + x^n + 1 = 0</math></p> |

### Question 13

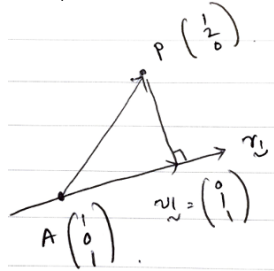
|     |                                                                                                                                                                                                                                                                                |                                                                                                                        |                                                                                                                     |
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| (a) | $\sum_{i=1}^n \sqrt{i} > \frac{2n\sqrt{n}}{3}$ <p>Step 1:<br/> For n = 1,<br/> <math display="block">\sqrt{1} &gt; \frac{2 \times \sqrt{1}}{3}</math> <math display="block">1 &gt; \frac{2}{3} \quad \text{True for } n = 1 \quad \checkmark</math></p> <p>Inductive step:</p> | <p>1 mark: proves basic step</p> <p>2 mark: completely correct proof for the inductive step and gives a conclusion</p> | <p>Most students received 2 out of 3 for this question. Inductive step was not shown clearly with correct logic</p> |
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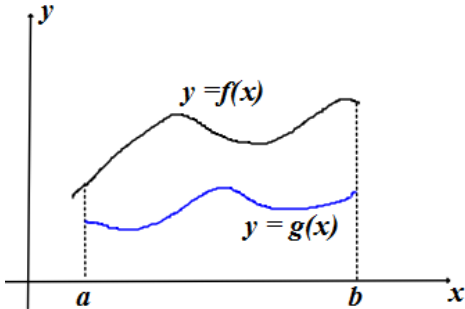
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                     |  |
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|     | <p>Suppose the result is true for <math>n = k</math>, then we prove it is true for <math>n = k+1</math></p> <p>Thus,</p> $\sum_{i=1}^k \sqrt{i} > \frac{2k\sqrt{k}}{3}, \text{ prove}$ $\sum_{i=1}^{k+1} \sqrt{i} > \frac{2(k+1)\sqrt{k+1}}{3}$ $\sum_{i=1}^{k+1} \sqrt{i} = \sum_{i=1}^k \sqrt{i} + \sqrt{k+1}$ $\geq \frac{2k\sqrt{k}}{3} + \sqrt{k+1}$ <p>We'll prove that</p> $\frac{2k\sqrt{k}}{3} + \sqrt{k+1} > \frac{2(k+1)\sqrt{k+1}}{3}$ $\frac{2k\sqrt{k} + 3\sqrt{k+1}}{3} > \frac{2(k+1)\sqrt{k+1}}{3}$ $2k\sqrt{k} + 3\sqrt{k+1} > 2(k+1)\sqrt{k+1}$ $2k\sqrt{k} > (2k-1)\sqrt{k+1}$ $4k^2k > (2k-1)^2(k+1)$ $4k^3 > (4k^2 - 4k + 1)(k+1)$ $4k^3 > (4k^3 - 3k + 1)$ $0 > -3k + 1 \quad \forall k \geq 1 \quad k \in \mathbb{Z}^+ \quad \checkmark \quad \checkmark$ <p>Thus the result is true for <math>n = k</math>, then it is true for <math>n = k+1</math></p> <p>Hence by principle of mathematical induction, true for all <math>n \geq 1</math></p> | <p>1 mark: progress toward the inductive proof</p>                                                                                                                                                                                                                  |  |
| (b) | <p>Let ABCD be the parallelogram</p> $\vec{AB} = \vec{DC} = \vec{a}$ $\vec{AD} = \vec{BC} = \vec{b}$ <p>Let <math>\vec{AF} = x\vec{AC}</math> and <math>\vec{EF} = y\vec{ED}</math></p> $\vec{AE} = \frac{\vec{a}}{2}, \quad \vec{ED} = \vec{b} - \frac{\vec{a}}{2}, \quad \vec{EF} = y\left(\vec{b} - \frac{\vec{a}}{2}\right),$ $\vec{AF} = \frac{\vec{a}}{2} + y\left(\vec{b} - \frac{\vec{a}}{2}\right)$ $= x(\vec{a} + \vec{b}) \quad \checkmark$ $x(\vec{a} + \vec{b}) = \frac{\vec{a}}{2} + y\left(\vec{b} - \frac{\vec{a}}{2}\right) \quad \checkmark$ <p>Equating coefficients of <math>\vec{a}</math> and <math>\vec{b}</math>, we get</p> $x = \frac{1}{2}(1 - y) \text{ and } x = y$                                                                                                                                                                                                                                                                          |  <p>1 mark: Writes <math>\vec{AF} = x(\vec{a} + \vec{b})</math></p> <p>1 mark: writes <math>\vec{AF} = \frac{\vec{a}}{2} + y\left(\vec{b} - \frac{\vec{a}}{2}\right)</math></p> |  |

|  |                                                                                                                                                                                                              |                                                                                                                                |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>Thus,</p> $y = \frac{1}{2}(1 - y)$ $y = \frac{1}{3} = x$ <p>Hence,</p> $\vec{AF} = \frac{1}{3} \vec{AC}; \quad \vec{EF} = \frac{1}{3} \vec{ED};$ $AF:FC = 1:2$ $EF:FD = 1:2 \quad \quad \quad \checkmark$ | <p>1 mark: equates the coefficients of <math>\vec{a}</math> and <math>\vec{b}</math>, and evaluates y and hence the result</p> |  |
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|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                               |              |
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| (c)(i)      | $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 2x^3 - 10x$ $\therefore \frac{1}{2}v^2 = \frac{x^4}{2} - 5x^2 + c$ <p style="text-align: center;"><b>1 mark</b></p> <p>When <math>x = 0</math>, <math>v = u \rightarrow c = \frac{1}{2}u^2</math></p> $\therefore \frac{1}{2}v^2 = \frac{x^4}{2} - 5x^2 + \frac{1}{2}u^2$ <p style="text-align: center;"><b>1 mark</b> <span style="float: right;">☑</span></p> $v^2 - u^2 = x^4 - 10x^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Applies <math>\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)</math> and integrates</p> <p>1 mark: correctly evaluates the constant of integration and proves the result</p>                                                                                                                                            | Well done    |
| (ii)        | <p>If <math>u = 3</math> then <math>v^2 - 9 = x^4 - 10x^2</math></p> $v^2 = x^4 - 10x^2 + 9$ $= (x^2 - 9)(x^2 - 1)$ $= (x - 1)(x + 1)(x - 3)(x + 3) \quad \text{1 mark}$  <p>Since, <math>v^2 \geq 0</math> for motion to exist then <math>(x - 1)(x + 1)(x - 3)(x + 3) \geq 0</math> and since the particle starts at <math>x = 0</math> with <math>v = 3</math> it is moving to the right.</p> <p style="text-align: center;"><b>1 mark</b></p> <p>At <math>x = 1</math>, <math>v = 0</math> and <math>\ddot{x} = -8</math> and so the particle will move to the left until it reaches <math>x = -1</math> where <math>v = 0</math> and <math>\ddot{x} = 8</math>. This means the particle will then move to the right until <math>v = 0</math> again at <math>x = 1</math>.</p> <p style="text-align: center;"><b>1 mark</b></p> <p>Thus it oscillates in the interval <math>-1 \leq x \leq 1</math></p> | <p>1 mark: correctly factorises and solves the inequality</p> <p>2 marks: uses the signs of <math>\dot{x}</math> and <math>\ddot{x}</math> to describe the direction of motion and proves the particle oscillates in the interval <math>-1 \leq x \leq 1</math></p> <p>1 mark: describes the motion at least at one point</p> | Poorly done. |
| (iii)       | Not SHM since $\ddot{x} \neq -n^2(x - b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1 mark                                                                                                                                                                                                                                                                                                                        |              |
| Question 14 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                               |              |
| a)(i)       | $(a - b)^2 > 0 \quad (a \neq b)$ $\therefore a^2 - 2ab + b^2 > 0 \quad \text{☑}$ $\therefore a^2 + b^2 > 2ab \text{ ----- } \boxed{1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1 mark: proves the result                                                                                                                                                                                                                                                                                                     | Well done    |
| a)(ii)      | <p>Similarly <math>b^2 + c^2 &gt; 2bc \text{ ----- } \boxed{2}</math></p> <p>and <math>c^2 + a^2 &gt; 2ca \text{ ----- } \boxed{3} \quad \text{☑}</math></p> <p>Now <math>\boxed{1} + \boxed{2} + \boxed{3}</math> gives</p> $2(a^2 + b^2 + c^2) > 2(ab + bc + ca) \quad \text{☑}$ $\therefore a^2 + b^2 + c^2 > ab + bc + ca$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1 mark: writes the statements</p> <p>1 mark: adds the statements to prove the result</p>                                                                                                                                                                                                                                   | Well done    |

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                         |                                                                       |
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| a)(iii) | <p>Using the result in (ii)<br/> Let <math>a \rightarrow ab</math>; <math>b \rightarrow bc</math>; <math>c \rightarrow ca</math>.</p> $\therefore a^2b^2 + b^2c^2 + c^2a^2 > ab.bc + bc.ca + ca.ab \quad \checkmark$ $a^2b^2 + b^2c^2 + c^2a^2 > abc(a + b + c) \quad \checkmark$ $\therefore \frac{a^2b^2 + b^2c^2 + c^2a^2}{a+b+c} > abc$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>2 marks: proves the result</p> <p>1 mark: minor error in the proof</p>                                                                                                                               | Well done                                                             |
| (b)(i)  | $Z^n + Z^{-n}$ $= (\cos n\theta + i \sin n\theta) + (\cos(-n\theta) + i \sin(-n\theta))$ $= 2 \cos n\theta$ <p><b>1 mark</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 mark: proves the result                                                                                                                                                                               | Well done                                                             |
| (ii)    | $2f(\theta) = 2 + k(2\cos\theta) + k^2(2\cos 2\theta) + \dots + k^n(2\cos n\theta)$ $= 2 + k(z + z^{-1}) + k^2(z^2 + z^{-2}) + \dots + k^n(z^n + z^{-n}) + \dots$ $= (1 + kz + k^2z^2 + \dots + k^n z^n + \dots) + (1 + \frac{k}{z} + (\frac{k}{z})^2 + \dots + (\frac{k}{z})^n + \dots)$ <p>Thus,</p> $f(\theta) = \frac{1}{2} \left[ \frac{1}{1 - kz} + \frac{1}{1 - \frac{k}{z}} \right]$ $f(\theta) = \frac{1}{2} \left[ \frac{1}{1 - kz} + \frac{z}{z - k} \right] \quad \text{1 mark}$ $\frac{1}{1 - kz} = \frac{1}{1 - k\cos\theta - i k\sin\theta}$ $= \frac{1 - k\cos\theta + i k\sin\theta}{(1 - k\cos\theta)^2 + k^2 \sin^2 \theta}$ $= \frac{1 - k\cos\theta + i k\sin\theta}{1 - 2k\cos\theta + k^2}$ $\frac{z}{z - k} = \frac{(\cos\theta + i\sin\theta)}{(\cos\theta - k) + i\sin\theta}$ $= \frac{(\cos\theta + i\sin\theta)[(\cos\theta - k) - i\sin\theta]}{[(\cos\theta - k) + i\sin\theta][(\cos\theta - k) + i\sin\theta]}$ $= \frac{1 - k\cos\theta - i k\sin\theta}{1 - 2k\cos\theta + k^2}$ $f(\theta) = \frac{1}{2} \left[ \frac{1 - k\cos\theta + i k\sin\theta + 1 - k\cos\theta - i k\sin\theta}{1 - 2k\cos\theta + k^2} \right]$ $= \frac{1 - k\cos\theta}{1 - 2k\cos\theta + k^2}$ | <p>1 mark: applies the result from (i) and find the sum of the infinite series</p> <p>2 marks: Rationalises the denominator and finds the result</p> <p>1 mark: attempts to simplify the expression</p> | Some students experienced difficulty in rationalising the denominator |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                          |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (iii)       | <p>For <math> k  &lt; 1, \theta = \frac{\pi}{2}</math>,</p> $1 + k \cos \theta + k^2 \cos 2\theta + \dots + k^n \cos n\theta + \dots$ $= 1 - k^2 + k^4 - k^6 + \dots$ $= \frac{1}{1 - (-k^2)} = \frac{1}{1 + k^2}$ <p>Also,</p> $\frac{1 - k \cos \theta}{1 - 2k \cos \theta + k^2} = \frac{1 - k \cos \frac{\pi}{2}}{1 - 2k \cos \frac{\pi}{2} + k^2} = \frac{1}{1 + k^2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1 mark: substitutes and verifies the result                                                                                                                                                                                                                                                                                                                                                              | Verify means you need to show both results give you the same value when $\theta = \frac{\pi}{2}$                                                                                         |
| c(i)        | $\vec{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda \\ 1 + \lambda \end{pmatrix},$ $\vec{r}_2 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 + \mu \\ 3\mu \\ 2\mu \end{pmatrix}$ <p>For point of intersection to occur,</p> $\begin{pmatrix} 1 \\ \lambda \\ 1 + \lambda \end{pmatrix} = \begin{pmatrix} 2 + \mu \\ 3\mu \\ 2\mu \end{pmatrix}$ $2 + \mu = 1 \rightarrow \mu = -1$ $\lambda = 3\mu = -3$ <p>Substitute in <math>1 + \lambda = 2\mu \rightarrow 1 - 3 = 2 \times -1</math> True.<br/>Hence, the two lines intersect</p>                                                                                                                                                                                                                                                                                  | <p>1 mark: Equates the parametric equations, finds the values of <math>\lambda</math> and <math>\mu</math>.</p> <p>1 mark: Substitutes into the third equation to prove true and thus proving the intersection.</p>                                                                                                                                                                                      | Well done                                                                                                                                                                                |
| c(ii)       | <p>It is the angle between the direction vectors</p> $\vec{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \text{ and } \vec{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ $\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{ \vec{v}_1   \vec{v}_2 }$ $\vec{v}_1 \cdot \vec{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ $= 0 \times 1 + 1 \times 1 + 1 \times 2 = 3$ $\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{ \vec{v}_1   \vec{v}_2 } = \frac{3}{\sqrt{0+1+1} \sqrt{1+1+4}} = \frac{3}{\sqrt{2} \sqrt{6}} = \frac{3}{\sqrt{12}} = \frac{\sqrt{3}}{2}$ $\theta = 30^\circ$                                                                                                                                                                                                                                                                                                                                     | <p>2 mark: Correctly finds the angle between the direction cosines</p> <p>Award 1 mark: Applies to the correct formula, however error in calculation</p>                                                                                                                                                                                                                                                 | Angle between two vectors is the angle between the direction vectors, not between the points $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ |
| c(iii)      | <p>Line <math>\vec{r}_1</math> passes through the point <math>A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}</math> and let <math>P \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}</math></p> <p>Thus <math>\vec{AP} = \begin{pmatrix} 1-1 \\ 2-0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}</math></p> <p><math>\vec{AP} \cdot \vec{v}_1 = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0 + 2 - 1 = 1</math></p> <p><math>\vec{v}_1 \cdot \vec{v}_1 = 2</math></p> <p>Shortest distance = <math> \vec{AP} - \text{Proj}_{\vec{v}_1} \vec{AP} </math></p> $= \left  \vec{AP} - \frac{\vec{AP} \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 \right $ $= \left  \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right  = \left  \begin{pmatrix} 0 \\ \frac{3}{2} \\ -\frac{3}{2} \end{pmatrix} \right  = \frac{3}{2} \sqrt{1+1} = \frac{3}{\sqrt{2}}$ | <p>3 marks: correct answer from correct working</p> <p>2 marks: Applies to the shortest distance formula and minor error</p> <p>1 mark: Correctly finds <math>\vec{AP}</math> and finds <math>\vec{AP} \cdot \vec{v}_1</math><br/>Or<br/>Correctly calculates the <math>\text{Proj}_{\vec{v}_1} \vec{AP}</math></p>  | A diagram would really help. Many students attempted substituting into a formula not realising the meanings of the results they are finding                                              |
| Question 15 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                          |
| (a)         | As shown in the diagram $f(x) > g(x)$ for $[a, b]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                          |

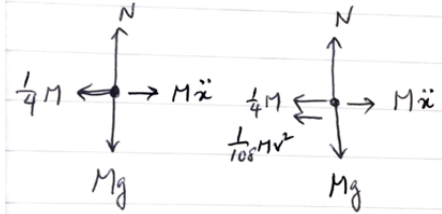
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                          |           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|         |  <p>Area under the curve <math>y = f(x)</math> is greater than area under the curve <math>y = g(x)</math> for <math>[a, b]</math>.<br/>Hence</p> $\int_a^b f(x) dx > \int_a^b g(x) dx$                                                                                                                                                                                                                                                                                                                           | <p>2 marks: proves the result providing valid explanation/working</p> <p>1 mark: attempts to explain the result, given a diagram or equivalent merit</p>                                                                                                                                                 | Well done |
| (b)(i)  | $I_n = \int_0^1 (1-t^2)^{\frac{n-1}{2}} dt,$ $I_n = \int_0^1 (1-t^2)^{\frac{n-1}{2}} \cdot 1 dt$ $= \left[ (1-t^2)^{\frac{n-1}{2}} \cdot t \right]_0^1 - \int_0^1 \left( \frac{n-1}{2} \right) (1-t^2)^{\frac{n-3}{2}} (-2t) \cdot t dt$ <p><b>1 mark</b></p> $= (n-1) \int_0^1 (1-t^2)^{\frac{n-3}{2}} \cdot t^2 dt$ $= (n-1) \int_0^1 (1-t^2)^{\frac{n-3}{2}} \cdot [1 - (1-t^2)] dt$ $= (n-1) \left\{ \int_0^1 (1-t^2)^{\frac{n-3}{2}} - (1-t^2)^{\frac{n-1}{2}} \right\} dt$ $I_n = (n-1) I_{n-2} - (n-1) I_n$ $(1+n-1)I_n = (n-1) I_{n-2}$ $\therefore n I_n = (n-1) I_{n-2} \quad n \geq 2$ | <p>1 mark: Applies integration by parts</p> <p>1 mark: manipulates to express the integrand to <math>(1-t^2)^{\frac{n-1}{2}}</math></p> <p>2 marks: correctly expresses the integrand in terms of <math>I_n</math> and <math>I_{n-1}</math> and gives the required result</p> <p>1 mark: Minor error</p> | Well done |
| (b)(ii) | <p><math>n I_n = (n-1) I_{n-2}</math> from (i)</p> <p>Similarly, <math>(n-1) I_{n-1} = (n-2) I_{n-3}</math></p> <p><math>(n-2) I_{n-2} = (n-3) I_{n-4}</math> <b>1 mark</b></p> <p>·</p> <p>·</p> <p>·</p> <p>Thus,</p> $J_n = n I_n I_{n-1}$ $= (n-1) I_{n-2} \times \frac{n-2}{n-1} I_{n-3}$ $= (n-2) I_{n-2} I_{n-3}$ <p><b>1 mark</b></p> <p>Summarising,</p> $J_n = n I_n I_{n-1} \quad n \geq 1$ $= (n-2) I_{n-2} I_{n-3}$                                                                                                                                                                  | <p>1 mark: Demonstrates the pattern</p> <p>1 mark: Demonstrates <math>J_n = n I_n I_{n-1} = (n-2) I_{n-2} I_{n-3}</math></p> <p>1 mark: proves <math>J_n = n I_n I_{n-1} = 2 I_2 I_1</math></p> <p>1 mark:</p>                                                                                           |           |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | $= (n-4)I_{n-4} I_{n-5}$ <p>.</p> <p>.</p> <p>.</p> $= [n - (n-2)]I_2 I_1$ $= 2 I_2 I_1 \quad \mathbf{1 \text{ mark}}$ $I_2 = \int_0^1 (1-t^2)^{\frac{1}{2}} dt = \frac{1}{4} \pi \times 1^2 = \frac{\pi}{4}$ $I_1 = \int_0^1 (1-t^2)^0 dt = 1$ <p>Thus</p> $J_n = 2 \times \frac{\pi}{4} \times 1 = \frac{\pi}{2} \quad \mathbf{1 \text{ mark}}$                                                                                                                                                                                                                                                                                                                                                                                 | Evaluates $I_2$ and $I_1$ and proves $J_n = \frac{\pi}{2}$                                                                                                                                                                                           |                                                                                                                                                                                                                                 |
| (b)(iii) | <p>For <math>0 \leq t \leq 1</math>,</p> $(1-t^2) \geq 0 \quad \mathbf{1 \text{ mark}}$ <p>Consider the functions</p> $(1-t^2)^{\frac{n-1}{2}} \text{ and } (1-t^2)^{\frac{n-3}{2}}$ $(1-t^2)^{\frac{n-1}{2}} - (1-t^2)^{\frac{n-3}{2}}$ $(1-t^2)^{\frac{n-3}{2}} [(1-t^2) - 1]$ $(1-t^2)^{\frac{n-3}{2}} [(-t^2)]$ <p>For <math>0 \leq t \leq 1</math>,</p> $(1-t^2)^{\frac{n-3}{2}} [(-t^2)] \leq 0 \quad (\text{must prove this})$ <p>Hence, <math>(1-t^2)^{\frac{n-1}{2}} &lt; (1-t^2)^{\frac{n-3}{2}} \quad \mathbf{1 \text{ mark}}</math></p> <p>Hence, using result in (a),</p> $\int_0^1 (1-t^2)^{\frac{n-1}{2}} dt < \int_0^1 (1-t^2)^{\frac{n-3}{2}} dt$ <p>Thus,</p> $0 < I_n < I_{n-1} \quad \mathbf{1 \text{ mark}}$ | <p>1 mark: explains <math>(1-t^2) \geq 0</math></p> <p>1 mark: proves <math>(1-t^2)^{\frac{n-1}{2}} &lt; (1-t^2)^{\frac{n-3}{2}}</math></p> <p>1 mark: refers to the result in (a) and explains why the inequality holds good.</p>                   | <p>You must prove the inequalities that you are using. The conditions that are given in the question must be addressed.</p> <p>You should also show reference to the results that you are using from any previous questions</p> |
| (b)(iv)  | <p>From (iii),</p> $I_n < I_{n-1} \quad \text{Similarly } I_{n+1} < I_n$ $I_{n+1} < I_n < I_{n-1}$ $I_n > 0$ <p>Thus</p> $I_{n+1} I_n < I_n I_n < I_n I_{n-1}$ <p>From (b) (ii),</p> $J_n = n I_n I_{n-1} = \frac{\pi}{2}$ $I_n I_{n-1} = \frac{\pi}{2n}$ $I_{n+1} I_n = \frac{\pi}{2(n+1)}$ <p>Thus,</p> $I_{n+1} I_n < (I_n)^2 < I_n I_{n-1}$ $\frac{\pi}{2(n+1)} < (I_n)^2 < \frac{\pi}{2n}$                                                                                                                                                                                                                                                                                                                                   | <p>2 marks: Uses the results in (b)(ii) and (b)(iii) to prove the result. Must show ALL lines of working</p> <p>1 mark: explains any of the results:</p> $I_n I_{n-1} = \frac{\pi}{2n}$ $I_{n+1} I_n = \frac{\pi}{2(n+1)}$ $I_{n+1} < I_n < I_{n-1}$ | Generally well done                                                                                                                                                                                                             |

$$\sqrt{\frac{\pi}{2n+2}} < I_n < \sqrt{\frac{\pi}{2n}}$$

### Question 16

(a)(i)



For  $0 \leq t \leq 20$ ,

$$M\ddot{x} = -\frac{1}{4}M$$

$$\ddot{x} = -\frac{1}{4}$$

For  $t > 20$ ,

$$M\ddot{x} = -\frac{1}{4}M - \frac{1}{108}Mv^2$$

$$\ddot{x} = -\frac{1}{108}(27 + v^2)$$

1 mark each : Draws the free body diagrams and writes the force equations

(ii)

$$\ddot{x} = -\frac{1}{4} \quad 0 \leq t \leq 20$$

$$\text{ie. } \frac{dv}{dt} = -\frac{1}{4}$$

$$\int_{60}^v dv = -\frac{1}{4} \int_0^t dt$$

$$v - 60 = -\frac{1}{4}t$$

$$v = -\frac{1}{4}t + 60 \quad \dots (1)$$

$$\frac{dx}{dt} = -\frac{1}{4}t + 60$$

$$\int_0^x dx = \int_0^t -\frac{1}{4}t + 60 dt$$

$$x = -\frac{1}{8}t^2 + 60t \quad \dots (2)$$

At  $t = 20$ , (1) and (2) give,

$$v = -\frac{1}{4} \times 20 + 60 = 55$$

$$x = -\frac{1}{8} \times 400 + 60 \times 20 = 1150 \text{ m}$$

2 mark: correctly proves the expression for  $v$  and  $x$  and substitutes to prove the results

1 mark: Correctly finds the expression for  $v$  or  $x$

(iii)

$$\ddot{x} = -\frac{1}{108}(27 + v^2) \quad t \geq 20$$

$$\text{ie. } v \frac{dv}{dx} = -\frac{1}{108}(27 + v^2)$$

$$\frac{v dv}{27 + v^2} = -\frac{1}{108} dx$$

$$\int_{55}^v \frac{v dv}{27 + v^2} = -\frac{1}{108} \int_{1150}^x dx$$

$$\left[ \frac{1}{2} \log(27 + v^2) \right]_{55}^v$$

$$= -\frac{1}{108} [x]_{1150}^x$$

2 mark: proves the result

1 mark:

Writes the expression

$$v \frac{dv}{dx} = -\frac{1}{108}(27 + v^2)$$

And attempts to prove the result

|      |                                                                                                                                                                                                                                                                                                                                                              |                                                     |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|      | $\frac{1}{2} \log \left[ \frac{27 + v^2}{27 + 55^2} \right] = -\frac{1}{108} (x - 1150)$ $\therefore x - 1150 = 54 \log \left[ \frac{27 + 55^2}{27 + v^2} \right]$ $x = 1150 + 54 \{ \ln(27 + (55)^2) - \ln(27 + v^2) \}$                                                                                                                                    |                                                     |
| (iv) | <p>The jet comes to rest when <math>v = 0</math>.<br/>         i.e.<br/> <math>x = 1150 + 54 \{ \ln(27 + (55)^2) - \ln(27 + v^2) \}</math><br/> <math>x = 1150 + 54 \{ \ln(27 + (55)^2) - \ln(27) \}</math><br/> <math>= 1405.296638 \dots</math><br/> <math>= 1405 \text{ m}</math><br/>         The jet comes to rest 1405 m from the touchdown point.</p> | 1 mark: Substitutes $v = 0$ and gets the $x$ value. |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                           |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| (bi)  | $1 - t + t^2 - t^3 + \dots + (-1)^{n-1} t^{n-1}$ <p>GP with <math>a = 1, r = -t</math> and <math>n</math> terms</p> $S_n = a \left( \frac{1 - (-t)^n}{1 - (-t)} \right)$ $= \frac{1}{1+t} - (-1)^n \frac{t^n}{1+t}$ <p>Thus,</p> $1 - t + t^2 - t^3 + \dots + (-1)^{n-1} t^{n-1} = \frac{1}{1+t} - (-1)^n \frac{t^n}{1+t}$ <p>Integrating both sides, 0 to <math>x</math>,</p> $\int_0^x 1 - t + t^2 - t^3 + \dots + (-1)^{n-1} t^{n-1} dt = \int_0^x \frac{1}{1+t} - (-1)^n \frac{t^n}{1+t} dt$ $\left[ t - \frac{t^2}{2} + \frac{t^3}{3} + \dots + (-1)^{n-1} \frac{t^n}{n} \right]_0^x = [\ln(1+t)]_0^x - \int_0^x (-1)^n \frac{t^n}{1+t} dt$ | <p>1 mark: finds the sum of GP correctly.</p> <p>1 mark: Integrates both sides of the equality 0 to <math>x</math>, proves the result</p> |
|       | <p>For <math>0 &lt; t &lt; x</math>,</p> $1 < 1+t$ <p>Hence, <math>\frac{1}{1+t} &lt; 1</math> <b>1 mark</b></p> <p>Also, <math>\frac{t^n}{1+t} &lt; t^n</math></p> <p>Thus</p> $\int_0^x \frac{t^n}{1+t} dt < \int_0^x t^n dt$ <b>1 mark</b>                                                                                                                                                                                                                                                                                                                                                                                                    | <p>1 mark: proves the inequality <math>\frac{1}{1+t} &lt; 1</math></p> <p>1 mark: proves the result</p>                                   |
| (iii) | $\int_0^x \frac{t^n}{1+t} dt < \int_0^x t^n dt = \left[ \frac{t^{n+1}}{n+1} \right]_0^x = \frac{x^{n+1}}{n+1}$ <p>Thus for <math>0 &lt; x &lt; 1</math>,</p> <p>Hence</p> $0 < \frac{x^{n+1}}{n+1} < \frac{1^{n+1}}{n+1} = \frac{1}{n+1}$ $\int_0^x \frac{t^n}{1+t} dt < \frac{1}{n+1}$                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                           |
| (iv)  | <p>From (i),</p> $x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + (-1)^n \frac{x^n}{n} = \log_e(1+x) - (-1)^n \int_0^x \frac{t^n}{1+t} dt$ <p>From (ii), when <math>n \rightarrow \infty</math>,</p> $\int_0^x \frac{t^n}{1+t} dt < \frac{1}{n+1} \rightarrow 0$ <p>Thus as <math>n \rightarrow \infty</math>,</p> $x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + (-1)^n \frac{x^n}{n} + \dots = \log_e(1+x)$                                                                                                                                                                                                                                          | <p>2 marks: correct proof with references to previous results and explanation</p> <p>1 mark: minor error</p>                              |